SVM · GEOMETRY

# Distance and Margin Width in SVM

## Perpendicular distance, the two margin boundaries, and the hard-margin objective

From the point-to-line formula to 2 / ||w||

Where the 2 in the margin comes from, why the final +1 appears in the numerator,
and why maximizing the margin means minimizing ||w||.

# Three ideas carry the whole derivation

Each later slide expands one of these statements

**1**

## SVM distance is the ordinary point-to-line distance

Substitute the point into the boundary expression, take the absolute value, divide by $||w||$.

**2**

## The full margin equals 2 / ||w||

$H_+$ and $H_-$ sit at +1 and −1; each is 1 / $||w||$ from the decision boundary.

**3**

## Maximizing the margin is minimizing ½||w||²

The ±1 scaling removes the freedom to rescale w and b, so the norm has geometric meaning.

**Margin width = 2 / ||w|| → hard-margin SVM minimizes ½||w||²**

# Four terms are needed before any formula

Definitions used throughout the deck, stated for a linear SVM

## Hyperplane

**$w^T x + b = 0$**

A flat boundary that splits the feature space into two sides. A line in 2-D, a plane in 3-D.

## Normal vector w

**perpendicular to the boundary**

Sets the boundary's orientation. Its length $||w||$ scales every value of $w^T x + b$.

## Margin

**empty band around the boundary**

The distance between the two parallel boundaries $H_+$ and $H_-$ that bracket the data.

## Support vectors

**points on $H_+$ or $H_-$**

The training points that touch the margin and fix where it lies.

# Distance is the line expression evaluated at the point

Point P = ($x_0$, $y_0$) and line ax + by + c = 0, with a and b not both zero

$$d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$$

## Numerator

- Substitute the point into the whole left side of the line equation
- Take the absolute value: distance is never negative

## Denominator

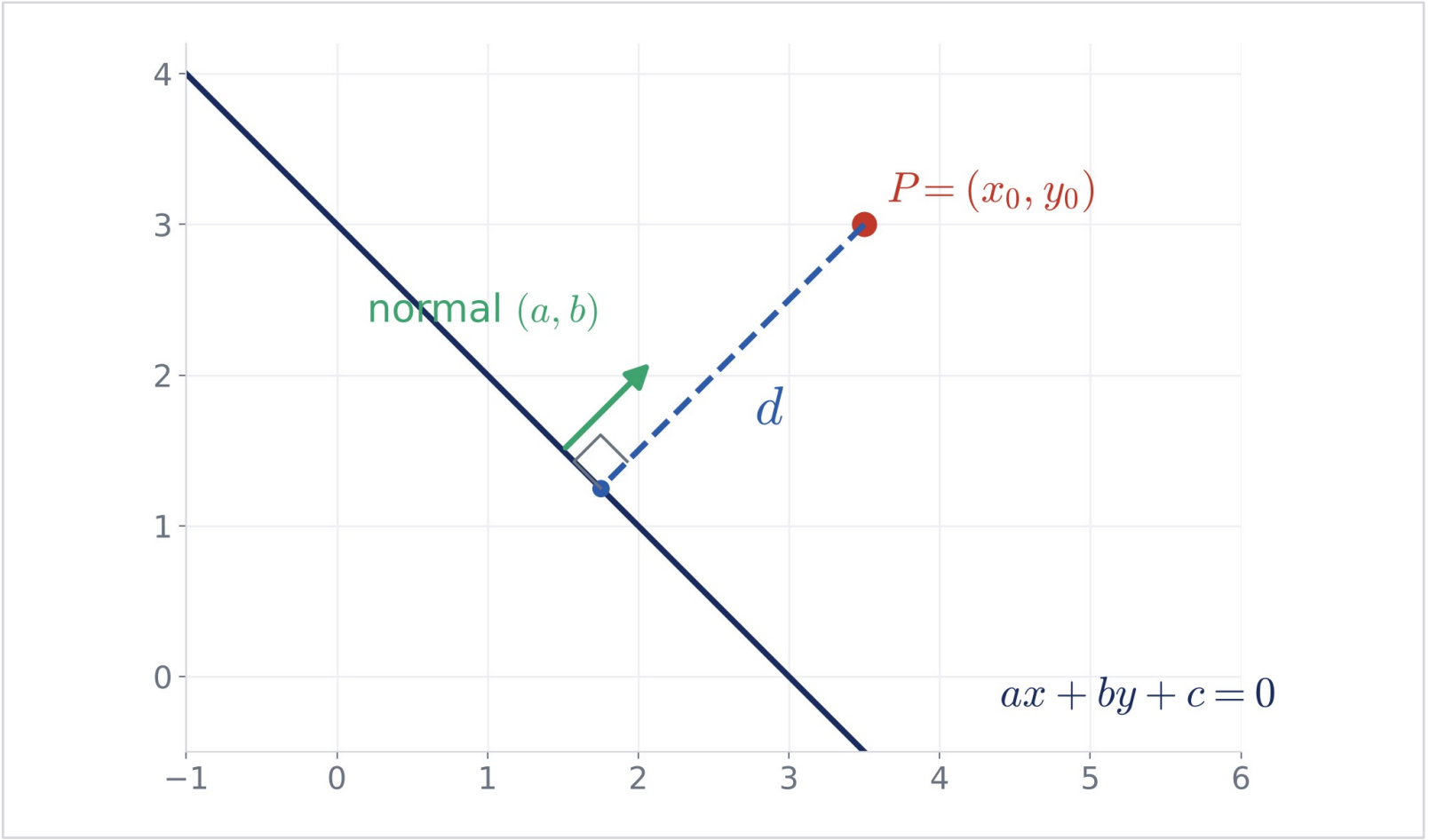
- $\sqrt{}(a^2 + b^2)$ is the length of the normal (a, b)
- It removes the effect of rescaling the coefficients

# The shortest path runs along the normal

The dashed segment meets the line at a right angle

## How to read

- Red point: P
- Green arrow: normal (a, b)
- Dashed segment: d
- Right angle at the foot



# The SVM distance is the same formula in vector form

Decision boundary $w^T x + b = 0$; w is the normal vector and b the intercept

$$d(\mathbf{x},\ \text{boundary}) = \frac{|\mathbf{w}^\top \mathbf{x} + b|}{\|\mathbf{w}\|}$$

$$\mathbf{w} = (w_1, w_2), \qquad \|\mathbf{w}\| = \sqrt{w_1^2 + w_2^2}$$

## Mapping

- (a, b) becomes w, c becomes b

## Scope

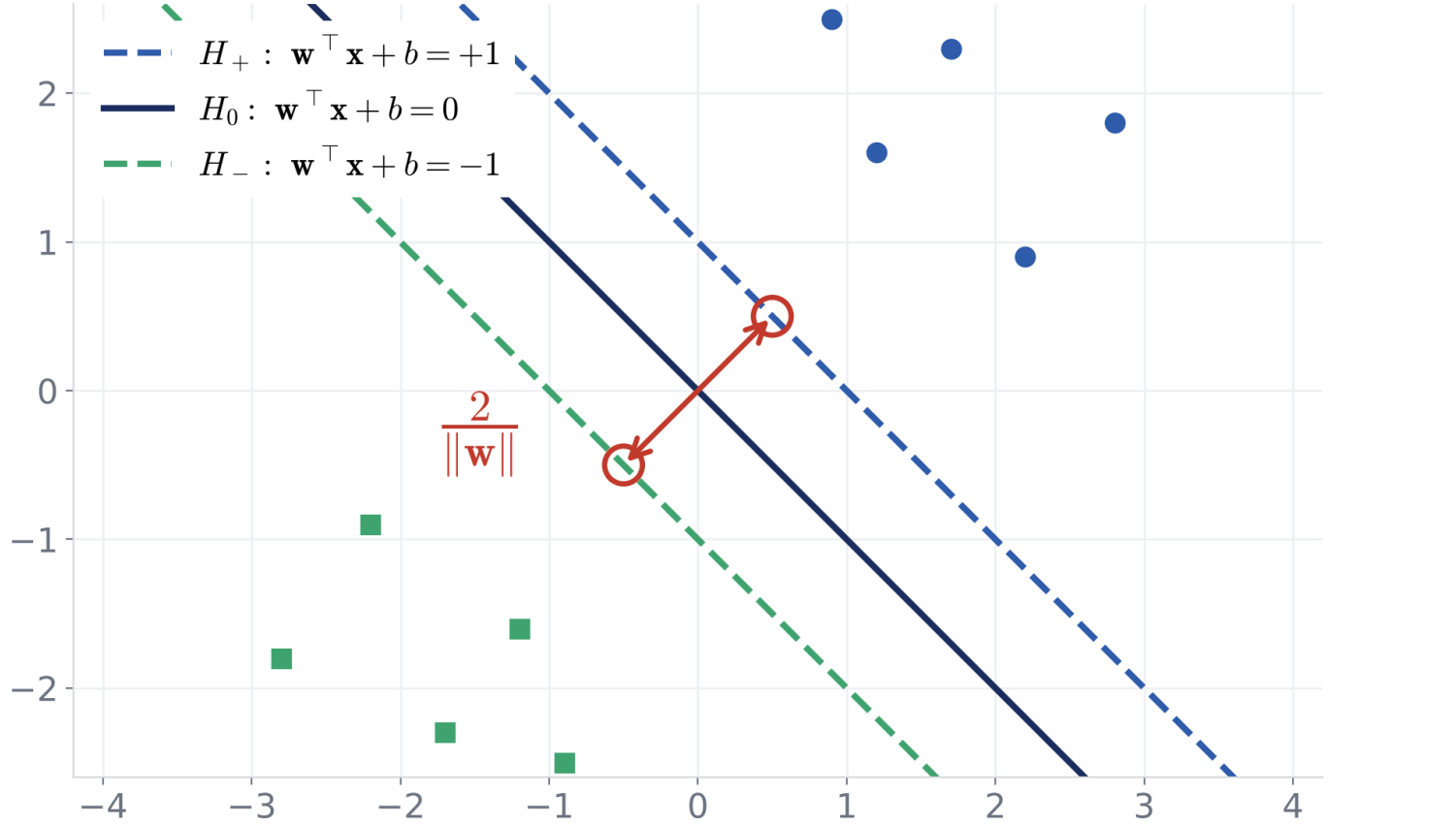
- Holds for hyperplanes in any dimension

# Margin boundaries sit on both sides of $H_0$

Standard SVM scaling puts them at $w^T x + b = +1$ and $-1$

## How to read

- Solid line: $H_0$
- Dashed lines: $H_+$, $H_-$
- Circled points: support vectors
- Red arrow: margin



# The full margin equals 2 / ||w||

Each margin boundary is 1 / ||w|| from the central decision boundary

$$H_+ : \ \mathbf{w}^\top \mathbf{x} + b = +1$$
$$H_0 : \ \mathbf{w}^\top \mathbf{x} + b = 0$$
$$H_- : \ \mathbf{w}^\top \mathbf{x} + b = -1$$

$$\underbrace{\frac{1}{\|\mathbf{w}\|}}_{H_+ \to H_0} + \underbrace{\frac{1}{\|\mathbf{w}\|}}_{H_0 \to H_-} = \frac{2}{\|\mathbf{w}\|}$$

Next slide: the same 2 / ||w|| obtained in a single distance calculation

# One distance calculation gives the whole margin

Take a point $x_+$ on $H_+$ and measure its distance to $H_-$

$$(1) \quad \mathbf{w}^\top \mathbf{x}_+ + b = 1$$

$$(2) \quad \mathbf{w}^\top \mathbf{x} + b = -1 \implies \mathbf{w}^\top \mathbf{x} + b + 1 = 0$$

$$(3) \quad d = \frac{|\mathbf{w}^\top \mathbf{x}_+ + b + 1|}{\|\mathbf{w}\|} = \frac{|1 + 1|}{\|\mathbf{w}\|} = \frac{2}{\|\mathbf{w}\|}$$

## Key move

- Rewrite $H_-$ with zero on the right side before substituting $x_+$

# The two 1s in the numerator have different origins

The numerator is still the line expression evaluated at a point

$$d = \frac{\left| \underbrace{\mathbf{w}^\top \mathbf{x}_+ + b}_{=1 \text{ (point on } H_+)} \quad \underbrace{+1}_{\text{from rewriting } H_-} \right|}{\|\mathbf{w}\|} = \frac{2}{\|\mathbf{w}\|}$$

## First 1

- $w^T x + b$ evaluated at the chosen point $x_+$
- It equals 1 because $x_+$ lies on $H_+$

## Final +1

- Moving $H_-$'s right side, −1, to the left
- It belongs to the target line, not the point

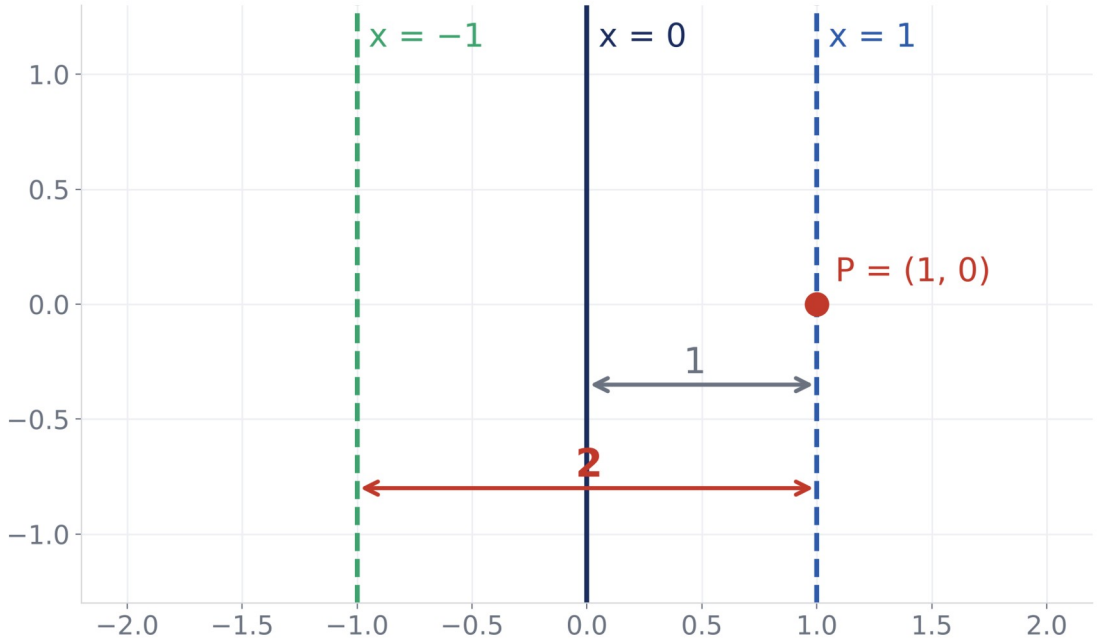
# A point on one margin is 2 units from the other

Boundary x = 0, margins x = 1 and x = −1, point P = (1, 0)

$$d = \frac{|1 \cdot 1 + 0 \cdot 0 + 1|}{\sqrt{1^2 + 0^2}} = \frac{2}{1} = 2$$

## Steps

- Rewrite x = −1 as x + 1 = 0
- Read a = 1, b = 0, c = 1
- Substitute P = (1, 0)
- Distance to x = 0 is 1



# Maximizing 2 / ||w|| is minimizing ½||w||²

Linearly separable data; labels $y_i$ are −1 or +1

$$y_i \left(\mathbf{w}^\top \mathbf{x}_i + b\right) \geq 1, \qquad y_i \in \{-1, +1\}$$

$$\max_{\mathbf{w},b} \frac{2}{\|\mathbf{w}\|} \iff \min_{\mathbf{w},b} \frac{1}{2}\|\mathbf{w}\|^2$$

## Constraint

- Every point lies on or outside its margin

## Why ½||w||²

- Same optimum, smooth and convex

# The ±1 scaling gives the norm its geometric meaning

Rescaling w and b by the same positive k leaves the boundary unchanged

$$(\mathbf{w},\, b) \;\to\; (k\mathbf{w},\, kb),\; k > 0: \qquad k\mathbf{w}^\top \mathbf{x} + kb = 0 \iff \mathbf{w}^\top \mathbf{x} + b = 0$$

## The problem

- Any k > 0 gives the same boundary
- Without a fix, ||w|| could be made arbitrarily small

## The fix

- Require the closest points to satisfy $y_i(w^\intercal x_i + b) = 1$
- Then ||w|| measures the margin: width = 2 / ||w||

# The max-margin reading holds only for hard margin

Soft-margin SVM trades margin width against a penalty for violations

| | Hard margin | Soft margin |
|---|---|---|
| Data | Linearly separable | May overlap |
| Constraint | $y_i(w^T x_i + b) \geq 1$ | Violations allowed |
| Objective | Minimize $\frac{1}{2}\|\|w\|\|^2$ | $\frac{1}{2}\|\|w\|\|^2$ plus a violation penalty |
| Margin | Exactly $2 / \|\|w\|\|$, empty | $2 / \|\|w\|\|$, some points inside |

# The margin formula follows from one distance rule

Point-to-line distance, ±1 scaling, and the norm of w

**1**

## Distance = |expression at the point| / length of the normal

$d = |w^T x + b| / ||w||$ is the point-to-line formula in vector form.

**2**

## Margin = 2 / ||w||

The first 1 comes from the point on $H_+$; the final +1 comes from rewriting $H_-$.

**3**

## Maximize margin ⇔ minimize ½||w||²

The ±1 scaling fixes the freedom to rescale (w, b); this reading holds for hard margin.

**Which line we measure to determines the constant in the numerator**